

# Shakoor Shaik

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## EDUCATION

### University of Toronto

Toronto, ON

*Bachelor of Science in Computer Science, Software Engineering — Dean's List 25'*

*April 2028 (Expected)*

**Courses:** Software Design (Object-Oriented Programming, Java), Systems Programming (Unix/Linux), Data Structures & Analysis, Computer Organization (Assembly), Intro to Machine Learning

## TECHNICAL SKILLS

**Languages:** Python, Java, C, Javascript/Typescript, SQL, HTML/CSS, R

**Frameworks:** React.js, Node.js, Express.js, p5.js, Flask, FastAPI, REST API, scikit-learn, pandas, NumPy, TensorFlow

**Technologies:** Git/Github, Firebase DB, Google Cloud Platform, PostgreSQL, Adobe Suite, Linux/Unix, Jira

**Developer Tools:** Visual Studio, Android Studio, IntelliJ, Eclipse, JUnit, Jupyter Notebook, R Studio, Pycharm

## EXPERIENCE

### STEM Instructor

Sep. 2025 – Present

*Exceed Robotics*

*Toronto, ON*

- Mentored **20+ students** in Python, C, and Arduino principles, debugging, and building projects for **12+ hrs/week**
- Led usage of Google Cloud API and AWS SDK with Raspberry Pi to develop AI image/audio detection apps

### Data Processing Engineer

Sep. 2024 – Sep. 2025

*University of Toronto Aerospace Team*

*Toronto, ON*

- Designed and trained **2D/3D CNN** models in Python for hyperspectral image classification on the FINCH satellite mission, collaborating with **5+ MSc** researchers on model architecture, data processing pipelines, and evaluation
- Corrected **40%** of existing errors in relational and NoSQL databases to enhance satellite capabilities
- **Contributed 1k+ lines** of python code to an established open-source codebase via Git and VS Code

### Full-Stack Engineer

Sep. 2024 – Dec. 2024

*CREATE UoFT*

*Toronto, ON*

- Built a full-stack web app via React and Django that facilitates smart course planning for **15000+** UoFT students
- Designed and deployed a Flask/Python backend with RESTful APIs, Jinja2 templating, and scalable MySQL DB
- Reduced deployment time **by 40%** via CI/CD automation, testing, and agile practices in a cross-functional team

## PROJECTS

### SMART Air | *Java, Android, Kotlin, Firebase Realtime Database, Firebase Auth, JUnit, XML*

- Built a scalable full-stack android health app serving **3** distinct user personas, utilizing MVP architecture to decouple business logic from the UI and ensure **100%** testability of core features using a Java/Kotlin codebase
- Optimized a NoSQL Firebase Realtime Database to implement secure authentication flows using Firebase Auth to manage multi-role access control, ensuring AODA/HIPAA-compliant data protection with sub-second data latency
- Accelerated feature delivery **by 30%** via optimized Git/GitHub workflows and Agile Scrum framework, improving sprint velocity and stakeholder communication

### Wine Quality Regression | *Python, scikit-learn, NumPy, pandas, PySR*

- Authored an academic research paper conducting a systematic comparison of 5 supervised ML regression models (KNN, LBR, BFR, MLP, Symbolic Regression) on the UCI Wine Quality Dataset (**6,497 samples**), documenting methodology, results, and findings across MSE, RMSE, MAE,  $R^2$ , and  $\text{Acc} \pm 1$  metrics
- Engineered a stratified data pipeline with StandardScaler normalization and 5-fold cross-validation to handle class imbalance, with Local Bayesian Regression achieving best performance (MSE: 0.351,  $R^2$ : 0.523,  $\text{Acc} \pm 1$ : 97.0%)
- Optimized hyperparameters across **1,650+** MLP configurations and tuned KNN/LBR neighborhood sizes, RBF kernel widths, and evolutionary SR expression trees using grid search and CV-MSE minimization

### Clippy | *Javascript, p5.js, Node.js, Express.js, HTML/CSS*

- **Won 2nd Place** at MUES Hackathon for building a modern AI-powered drawing platform inspired by MS Paint
- Engineered a custom graphics engine using p5.js to support dynamic multi-layer rendering and independent Undo/Redo history stacks
- Developed a scalable Node.js/Express backend to orchestrate asynchronous OpenAI API requests, securing API keys and delivering infinite, randomized drawing challenges with **50ms latency**